

Auteur: Xander De Blauwer

Studierichting: Informatica

Oefeningenreeks 4A nr. 62.4

$$\int \frac{x^2}{\sqrt{1-x^2}} dx$$

$$\text{Stel } x = \sin(t) \Rightarrow dx = \cos(t)dt$$

$$= \int \frac{\sin^2(t)}{\cos(t)} \cos(t)dt$$

$$= \int \sin^2(t)dt$$

$$= \int \frac{1 - \cos(2t)}{2} dt = \frac{1}{2}t - \frac{1}{4}\sin(2t) + c$$

$$= \frac{1}{2}\text{Bgsin}(x) - \frac{1}{2}\sin(t)\cos(t) + c$$

$$= \frac{1}{2}\text{Bgsin}(x) - \frac{1}{2}x\sqrt{1-x^2} + c$$

References